

# **Design and FPGA Prototyping of a LED Driver for Aircraft Cabin Lighting with Open-Source Tools**

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Faculty of Electrical Engineering and Computer Science

# **Design and FPGA Prototyping of a LED Driver for Aircraft Cabin Lightning with Open-Source Tools**

by

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## Table of contents

|                                             |           |
|---------------------------------------------|-----------|
| <b>Abstract</b>                             | <b>4</b>  |
| <b>Acknowledgements</b>                     | <b>5</b>  |
| <b>1. Introduction</b>                      | <b>6</b>  |
| 1.1. Scope of the Thesis . . . . .          | 6         |
| 1.2. Motivation and Challenges . . . . .    | 7         |
| 1.3. Methodological Contributions . . . . . | 7         |
| <b>2. Requirements</b>                      | <b>8</b>  |
| 2.1. Topology Selection: . . . . .          | 8         |
| 2.2. Block Diagram . . . . .                | 8         |
| <b>Bibliography</b>                         | <b>9</b>  |
| <b>Appendices</b>                           | <b>11</b> |
| <b>A. Tools</b>                             | <b>11</b> |
| <b>B. USB-Stick and Script Referencing</b>  | <b>12</b> |

## List of Figures

## List of Tables

## List of Abbreviations

**AC** — Alternating Current  
**AVDD** — Analog Supply Voltage  
**AVSS** — Analog Ground  
**CACE** — Computer-Aided Circuit Evaluation  
**CC** — Coupling Capacitor  
**C** — Capacitance  
 $C_{gd}$  — Gate–Drain Capacitance  
 $C_{gs}$  — Gate–Source Capacitance  
**CMOS** — Complementary Metal–Oxide–Semiconductor  
**CMRR** — Common-Mode Rejection Ratio  
**DC** — Direct Current  
**DFM** — Design for Manufacturability  
**DRC** — Design Rule Check  
**EDA** — Electronic Design Automation  
**EDA-OS** — Open-Source Electronic Design Automation  
**ESD** — Electrostatic Discharge  
**FF** — Fast–Fast Process Corner  
**FS** — Fast–Slow Process Corner  
**GDS / GDSII** — Graphic Data System (Final Layout Format)  
 $g_m$  — Transconductance  
**Hz** — Hertz  
**I/O** — Input/Output  
**IC** — Integrated Circuit  
 $I_d$  — Drain Current  
**IHP** — Innovations for High Performance Microelectronics  
**LED** — Light-Emitting Diode  
**LDO** — Low-Dropout Regulator  
**LHS** — Left-Hand Side  
**L** — Channel Length  
 $L_{min}$  — Minimum Channel Length  
**MC** — Monte-Carlo Simulation  
**MIM** — Metal–Insulator–Metal Capacitor  
**MOSFET** — Metal–Oxide–Semiconductor Field-Effect Transistor  
**MPW** — Multi-Project Wafer  
**μm** — Micrometre (Micron)  
 $n_g$  — Number of gates  
 $N_f$  — Number of fingers  
**NMOS** — n-channel MOSFET  
**nm** — Nanometre  
**OTA** — Operational Transconductance Amplifier  
**PAD** — Bond Pad  
**PCB** — Printed Circuit Board  
**Pcell** — Parameterized Cell

**PDK** — Process Design Kit  
**PEX** — Parasitic Extraction  
**PMIC** — Power Management Integrated Circuit  
**PMOS** — p-channel MOSFET  
**PPV** — Post-Processing Verification  
**PSRR** — Power Supply Rejection Ratio  
**PVT** — Process, Voltage, Temperature  
**Qt** — Qt Framework (used by KLayout GUI)  
**R** — Resistance  
 $R_{sub}$  — Substrate Resistance  
 $R_{well}$  — Well Resistance  
**SG13G2** — IHP 130-nm SiGe BiCMOS Technology  
**SME** — Small and Medium-Sized Enterprise  
**SS** — Slow–Slow Process Corner  
**SF** — Slow–Fast Process Corner  
**STI** — Shallow Trench Isolation  
**TT** — Typical–Typical Process Corner  
**UGB** — Unity-Gain Bandwidth  
**V** — Volt  
 $V_{DS}$  — Drain–Source Voltage  
 $V_{GS}$  — Gate–Source Voltage  
 $V_{IN}^+$  — Non-inverting Input Voltage  
 $V_{IN}^-$  — Inverting Input Voltage  
 $V_{OUT}$  — Output Voltage  
 $V_{REF}$  — Reference Voltage  
**W** — Channel Width  
**W/L** — Width-to-Length Ratio

## Declaration of Authorship

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## Abstract

## Acknowledgements

I would like to express my sincere gratitude to my dog, Chuckles, for eating my research notes multiple times. If it wasn't for you, I would have finished this thesis earlier.

# Chapter 1.

## Introduction

In recent years, the **open-source hardware ecosystem** has matured to a point where advanced *analog* and *mixed-signal* integrated circuit (IC) design has become increasingly feasible outside proprietary EDA environments. Building on this trend, the present work contributes to the **preparatory phase of a larger collaborative initiative** that aims to strengthen open-source-based chip design methodologies for small and medium-sized enterprises (SMEs), universities, and research institutes.

Using the design of an **LED driver IC** as a demonstrator, the overarching project seeks to **identify and close gaps** in the open-source design flow and to establish a reproducible, validated toolchain for mixed-signal system design.

### **i** Project Context

This thesis forms the **technical pre-work** of a multi-partner initiative that extends the *IIC-OSIC-TOOLS* environment with the *IHP SG13G2 OpenPDK*, enabling future open-source mixed-signal tapeouts for SMEs and universities.

The long-term objective of the parent project is to extend the **open-source IIC-OSIC-TOOLS** developed by *Prof. Pretl (JKU Linz)* with the **IHP OpenPDK SG13G2**, a 130 nm SiGe BiCMOS process offering analog and RF capabilities suited for power management, sensor, and communication ICs.

By integrating the PDK into the IIC-OSIC-TOOLS flow, the project aims to make **open-source chip design accessible for SMEs**, who have so far been restricted to board-level development due to the prohibitive cost and complexity of commercial EDA ecosystems. This initiative aligns with the global movement toward **democratizing IC design**, exemplified by the formation of the *IEEE SSCS Technical Committee on Open Source Ecosystem (TC-OSE)*.

### 1.1. Scope of the Thesis

Within this framework, the present thesis establishes the **proof of concept for the analog front-end** of the LED driver IC.

The work focuses on developing, simulating, and taping-out an **operational amplifier (OTA)** using the **IHP SG13G2 PDK** and the **IIC-OSIC-TOOLS**.

The design was submitted to the **IHP Open Multi-Project Wafer (MPW) Run in July 2025**, demonstrating a complete, fabrication-ready analog layout realized entirely with open-source tools such as **Xschem**, **Ngspice**, **KLayout**, **Klayout-PEX**, and **CACE**.

### **!** Technical Milestone

The successful **tape-out of the OTA** represents one of the earliest demonstrations of the SG13G2 OpenPDK using the IIC-OSIC toolchain and serves as a validation of the *open-source analog flow*.

## 1.2. Motivation and Challenges

The analog subsystem of an LED driver in **aircraft cabin lighting** must regulate voltages between **18 V and 32 V**, ensuring efficiency, low noise, and individual seat controllability.

Such power-management circuits are particularly sensitive to **layout-dependent parasitic effects**, which can degrade performance. Hence, a major emphasis of this work lies in the **DRC- and LVS-clean** design and its reliability in open-source flows.

## 1.3. Methodological Contributions

Beyond the circuit-level design, this research establishes a **validated analog verification workflow** under fully open conditions. The complete flow — from schematic capture, device sizing, and layout to LVS, PEX, and post-layout simulation — was developed within the *IIC-OSIC-TOOLS* environment.

Insights gained from the OTA design, including *device matching*, *guard-ring implementation*, *fill density*, and *PEX consistency*, are being integrated into improvements of the toolchain for future mixed-signal designs.

## Chapter 2.

### Requirements

A switch-mode LED driver IC needs to be designed to operate within an 18–36 V input range, support high-resolution (16-bit) PWM dimming for at least four RGBW channels with adjustable current control (1–40 mA), integrate protection features (ESD, over/under-voltage, over-temperature), and enable robust data communication with upstream and downstream ICs. The IC must also include an internal oscillator (25 MHz), internal generation of data line supply (3–5 VDC), and advanced diagnostics such as LED error detection.

#### 2.1. Topology Selection:

To meet the defined performance and protection requirements efficiently, a buck converter topology has been selected for the LED driver IC. This choice is based on the analysis and recommendations from Bernhard Wicht’s 2020 paper **Analog Building Blocks for DC-DC Converters**, [1] which highlights the buck converter as a suitable and energy-efficient architecture for step-down voltage regulation in integrated power management systems. Given the input voltage range of 18–36 V and the need for tightly regulated current control per channel (1–40 mA), the buck converter enables high efficiency and compact implementation. Its compatibility with mixed-signal integration and the ability to support precise current regulation and fast transient response makes it well aligned with the requirements of our LED driver IC, particularly in ensuring reliable performance across varying input conditions and thermal loads.

#### 2.2. Block Diagram

## Bibliography

- [1] B. Wicht, “Analog building blocks of dc-dc converters: Examining fundamental concepts,” *IEEE Solid-State Circuits Magazine*, vol. 12, no. 3, pp. 42–47, 2020, doi: [10.1109/mssc.2020.3002141](https://doi.org/10.1109/mssc.2020.3002141).



## Appendix A.

### Tools

This thesis was written using [Quarto](#)

## Appendix B.

### USB-Stick and Script Referencing

All directory structures and file paths presented in this appendix are intentionally simplified. They are provided to clarify the organization of the project and to indicate where specific scripts, netlists, verification outputs, and layout files can be found within the design environment.

```
thesis
  .gitattributes
  .gitignore
  _book/
  _quarto/
  _quarto.yml
  _bibliography
  index.qmd          # Main project page
  README.md
  references.qmd
  appendix           # Appendix
```